1. N4 iteration 2: double large primes (B2) and structured elimination (B1) vs the recorded B0

2026-10-21. Branch `claude/quartic-ic-n4-la-20261021`, protocol commit eab1bdfacd, records `results/`. Descriptive run report; interpretation in `README.md`. Cost in F_q-multiplication equivalents (M-eq); every IC unit verified kG = Q.

1.1. Result in brief

B2 verified on 51 of 51 units (first attempts retried and charged) and costs IC/rho = 0.024 at q = 1048507. Exponent on q >= 4051: B2 1.045 [1.037, 1.052], rho 1.476 [1.438, 1.508]; the recorded B0 baseline had 1.083. B1 (structured elimination) is a small stage gain and a regression at q = 262111. B2 is the new baseline. GHS-weak family only; plain rho; shared VM.

1.2. B2 against the matched rho

q	units ok	rho runs	B2 M-eq	rho M-eq	B2/rho w=0 (95%)	resolved
223	5/5	150	8.01e+05	2.08e+05	3.848 (2.665-5.251)	rho lower
499	5/5	150	8.23e+05	5.88e+05	1.400 (1.080-1.792)	unresolved
991	5/5	150	1.21e+06	1.53e+06	0.791 (0.710-0.881)	IC lower
2011	5/5	150	1.97e+06	4.54e+06	0.434 (0.388-0.485)	IC lower
4051	5/5	150	3.39e+06	1.32e+07	0.257 (0.231-0.288)	IC lower
8191	5/5	120	6.2e+06	4.07e+07	0.153 (0.139-0.168)	IC lower
16363	5/5	120	1.16e+07	1.13e+08	0.103 (0.094-0.114)	IC lower
32719	5/5	80	2.29e+07	2.8e+08	0.082 (0.074-0.091)	IC lower
65479	5/5	80	4.5e+07	9.2e+08	0.049 (0.043-0.055)	IC lower
131071	3/3	24	8.98e+07	2.55e+09	0.035 (0.030-0.042)	IC lower
262111	3/3	18	2.28e+08	6.4e+09	0.036 (0.029-0.046)	IC lower
524287	3/3	18	4.91e+08	2.01e+10	0.024 (0.019-0.033)	IC lower
1048507	3/3	12	1.09e+09	4.49e+10	0.024 (0.019-0.031)	IC lower

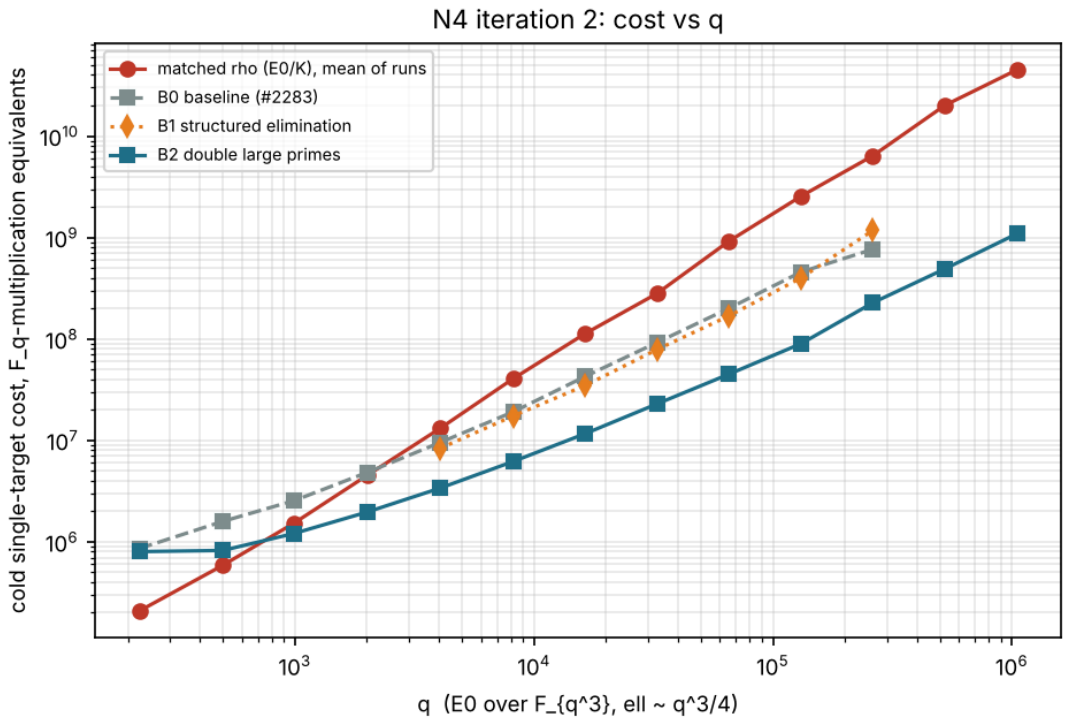


Figure 1: Cold cost vs q (recorded means). B0 and rho: records of PR 2283; B1, B2: this study.

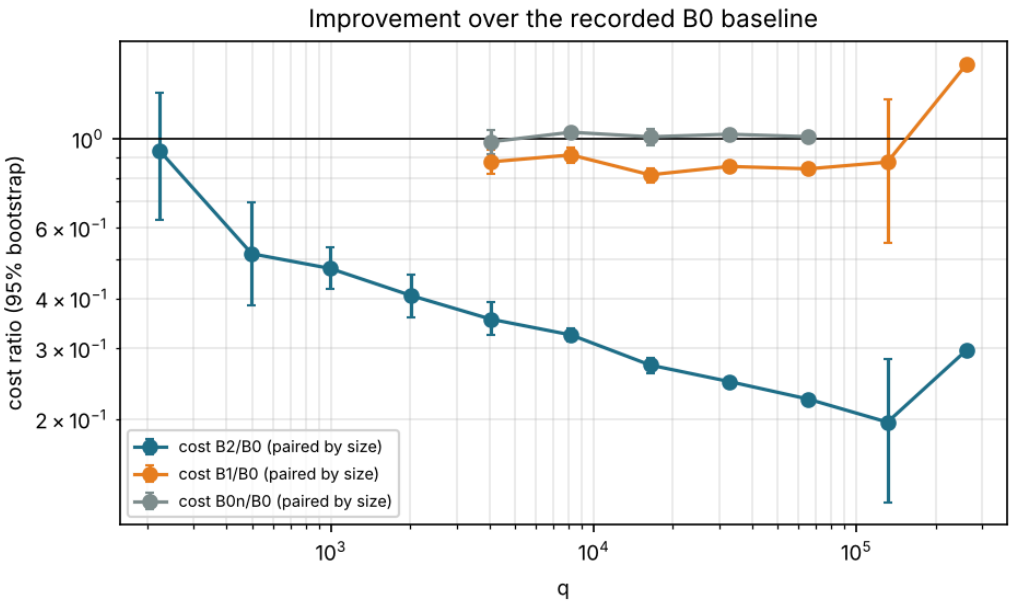


Figure 2: Cost ratios against the recorded B0 baseline, paired by size, 95% bootstrap over units.

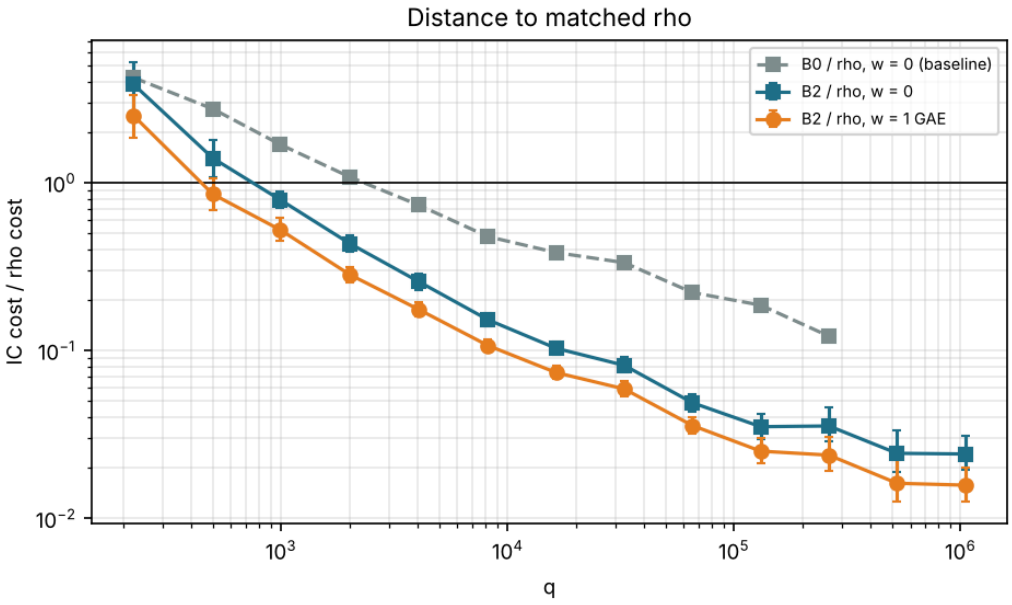


Figure 3: IC cost over matched rho cost with 95% bootstrap intervals.

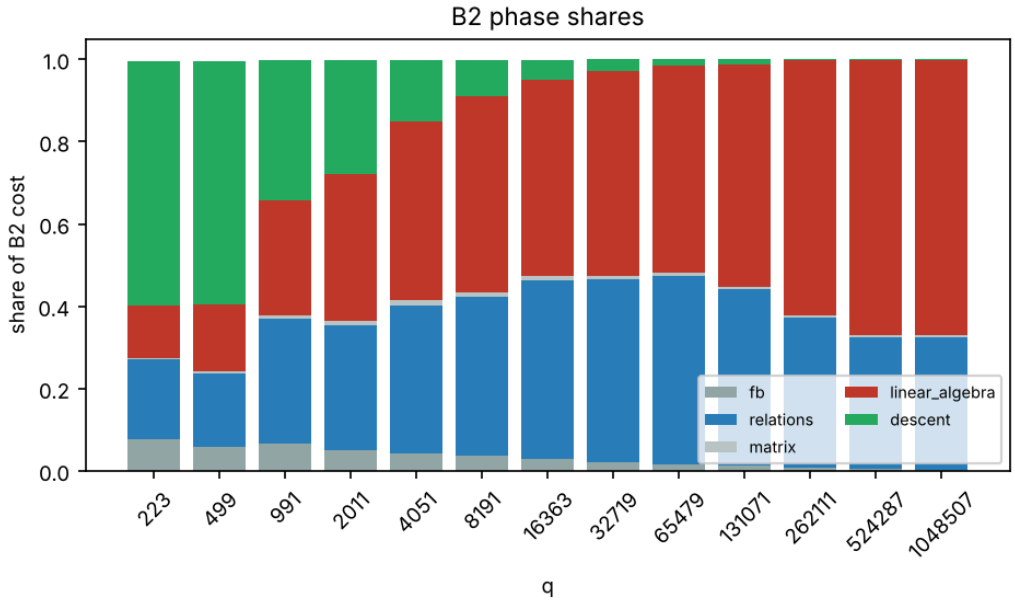


Figure 4: Phase shares of B2: linear algebra and relations; descent under 1% from q = 262111.

1.3. Paired comparison with B0

X/Y	q	X units	Y units	cost ratio (95%)
B0n/B0	4051	3	5	0.983 (0.919-1.049)
B0n/B0	8191	3	5	1.039 (1.016-1.062)
B0n/B0	16363	3	5	1.012 (0.965-1.059)
B0n/B0	32719	3	5	1.027 (1.007-1.046)
B0n/B0	65479	3	5	1.012 (1.005-1.019)
B1/B0	4051	5	5	0.876 (0.820-0.941)
B1/B0	8191	5	5	0.912 (0.871-0.949)
B1/B0	16363	5	5	0.813 (0.777-0.845)
B1/B0	32719	5	5	0.853 (0.837-0.870)
B1/B0	65479	5	5	0.842 (0.832-0.851)
B1/B0	131071	3	3	0.875 (0.550-1.254)
B1/B0	262111	3	3	1.534 (1.523-1.547)
B2/B0	223	5	5	0.932 (0.630-1.304)
B2/B0	499	5	5	0.517 (0.385-0.693)
B2/B0	991	5	5	0.475 (0.422-0.536)
B2/B0	2011	5	5	0.407 (0.359-0.459)
B2/B0	4051	5	5	0.355 (0.324-0.392)
B2/B0	8191	5	5	0.324 (0.313-0.336)
B2/B0	16363	5	5	0.273 (0.261-0.285)
B2/B0	32719	5	5	0.248 (0.245-0.252)
B2/B0	65479	5	5	0.224 (0.222-0.227)
B2/B0	131071	3	3	0.197 (0.124-0.282)
B2/B0	262111	3	3	0.297 (0.295-0.299)
B2/B1	4051	5	5	0.405 (0.374-0.441)
B2/B1	8191	5	5	0.356 (0.338-0.376)
B2/B1	16363	5	5	0.336 (0.324-0.349)
B2/B1	32719	5	5	0.291 (0.286-0.296)

B2/B1	65479	5	5	0.267 (0.263-0.270)
B2/B1	131071	3	3	0.225 (0.222-0.227)
B2/B1	262111	3	3	0.194 (0.192-0.196)

1.4. Exponents

range / arm	IC (w=0)	rho (w=0)	gap (w=0)
B2, q >= 4051	1.045 [1.037, 1.052]	1.476 [1.438, 1.508]	0.430 [0.391, 0.463]
B2, all 13 sizes	0.895 [0.878, 0.915]	1.482 [1.462, 1.499]	0.587 [0.558, 0.612]
B0 baseline, q >= 4051	1.083 [1.040, 1.129]	1.485 [1.437, 1.525]	0.402 [0.335, 0.468]

1.5. Limits

GHS-weak family only; plain rho without a negation map (up to sqrt 2); L0 shared VM; B2 first attempts at q = 223 included two wrong answers caught by the kG = Q check (vacuous hold-out at 30 columns); B1 regresses at q = 262111; not compared with an external implementation.